

Structural Shifts in AI Writing Bypass State-of-the-Art Detectors

UTS at ELOQUENT 2026 Voight-Kampff

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1. Motivation

The Voight-Kampff (VK) shared task, run jointly by ELOQUENT and PAN, asks whether a single text was written by a human or a language model [1, 2]. In 2024 the best-system fool rate against the PAN detector panel was **0.049**. In 2025 it jumped to **0.654**, driven by two prompting recipes that retrained nothing: literal back-translation plus a “be imperfect” instruction [3], and a nine-language rotation [4].

Research question. Which evasion attacks survive a deployer-class adversarial fine-tune that has already seen the 2025 attacks?

Main finding. An asymmetry. Attacks that push generations *out* of the detector’s training distribution (historical register, modernist narrative form) survive; attacks that pull them *into* it (mimicking the human training data) fail completely. The obvious counter-measure, retraining on real period prose, does not close the gap.

2. Method

Generators

Claude Opus 4.7, GPT-5.4, Gemini 2.5 Pro (closed frontier); Qwen3.6-35B-A3B, 4-bit (open MoE). All pinned to dated snapshots.

Attacks (13, in four families)

- 2025 replication:** vanilla, imperfection, roundtrip, roundtrip_imperf (PJS v2), lost_in_translation.
- Cross-decade register:** style_pre1923, style_translated_ru, style_l2_academic.
- Source-anchored, author-style, form:** synth_anchor (two-stage: generate a ~120-word period paragraph on an adjacent theme, then write the topic in its register); Borges / Sebald / Dillard imitations; stream_narrative.
- Bracketing:** Macko-pp retraining; real_anchor; pan_anchor (few-shot PAN mimicry).

Detectors

- In-distribution:** TF-IDF 1–4-gram SVM (PAN’25 baseline), RoBERTa-base fine-tuned on PAN’25 (val AUC 0.9994), Binoculars [5], LogPerplexity and LogRank on GPT-2M.
- Deployer-class (adversarial):** RoBERTa-large fine-tuned on our generations + PAN’25 humans, named after the 2025 PAN winner’s adversarial recipe [6]. **Macko-replica:** leave-one-genre-bucket-out. **Macko-LOSO:** leave-one-strategy-out. **Macko-pp:** Macko-LOSO + ~1,000 pre-1923 Project Gutenberg passages (PG-19) [7] on the human side.
- Neutral committee:** RoBERTa-large, RoBERTa-large + homograph augmentation, DeBERTa-v3-large, each trained on PAN’25 only.

Evaluation

Fool rate = fraction of generations scored $P(\text{LLM}) < 0.5$. 66 topics from VK 2024/25/26 ($n = 1,320$ for the 4×5 factorial); the 20 VK-2026 test topics are held out for the new attacks. Naturalness: cross-judge $P(\text{human})$ from three frontier LLMs, self-family judge dropped.

3. Mechanism

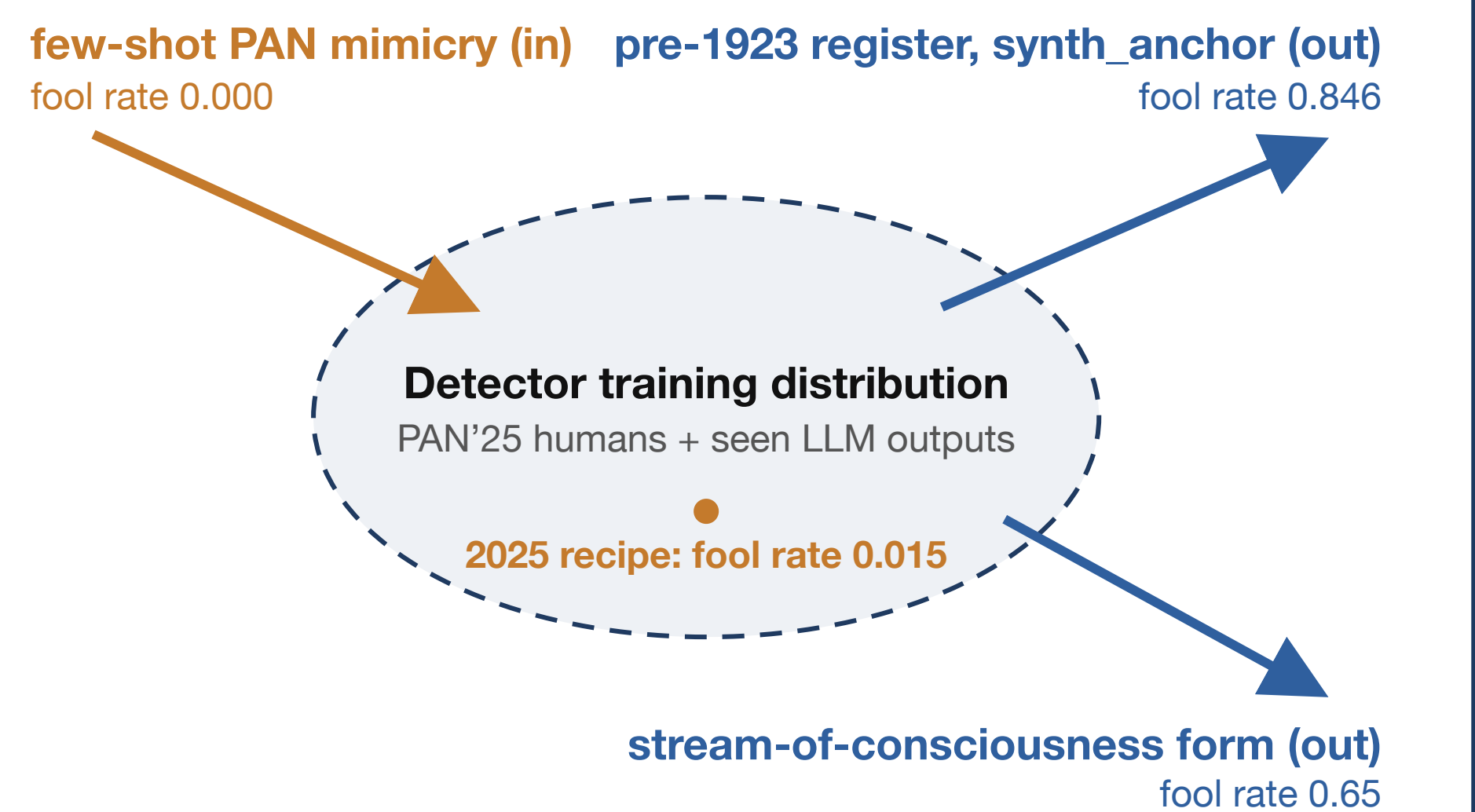


Figure 1. Out-of-distribution direction, not surface mimicry, defeats adversarial fine-tuning. Fool rates on Macko-pp (Macko-LOSO for the 2025 recipe), Claude Opus 4.7, 20 held-out VK-2026 topics.

4. Results

4.1 Adversarial fine-tuning closes the 2025 recipe

PAN’25 RoBERTa, the strongest in-distribution detector, is the *most* fooled by PJS v2: $P(\text{LLM}) \approx 1.00$ on vanilla text drops to ≈ 0.001 . One adversarial fold reverses this on every generator. The recipe becomes self-defeating: it makes text *more* detectable than vanilla.

Generator	RoBERTa-PAN’25 fool	Macko-replica fool	Δ vs vanilla
Claude Opus 4.7	0.99	0.025	−0.71
GPT-5.4	0.96	0.012	−0.52
Gemini 2.5 Pro	0.99	0.025	−0.74
Qwen3.6-35B-A3B	0.57	0.014	−0.50

Table 1. Fool rate under roundtrip_imperf (PJS v2); higher = better attack. Sign inversion significant at $\alpha = 0.05$ (paired bootstrap).

4.2 Cross-decade register attacks survive

Register-level perturbations (vocabulary epoch, translation register, L2 grammar) exit the training distribution further than grammatical perturbation. Predicted ordering pre-1923 > translated-Russian > L2-academic confirmed. synth_anchor dominates style_pre1923 on every detector column, reaching $\approx 50\times$ the 2025 recipe’s fool rate on the same detector, generator, and topics.

Strategy	RoBERTa	TF-IDF	Macko-replica	Macko-LOSO
roundtrip_imperf	0.98	0.92	0.05	0.015
style_pre1923	0.15	0.58	0.50	0.640
style_translated_ru	0.07	0.45	0.42	0.483
style_l2_academic	0.03	0.38	0.31	0.395
synth_anchor	0.57	0.76	0.70	0.798

Table 2. Fool rate, Claude Opus 4.7, 20 held-out VK-2026 topics.

4.3 The obvious counter-measure fails

Macko-pp adds ~1,000 pre-1923 Gutenberg passages to the human side. Prediction P11.1, formulated before training: synth_anchor fool rate falls from 0.798 to ≤ 0.20 . **Falsified:** it rises to 0.846. Controls stay caught, so the detector is not decalibrated; it is specifically blind to period register. A real Gutenberg anchor underperforms the synthetic one (0.778 vs 0.846), likely because the synthetic anchor is thematically matched to the topic.

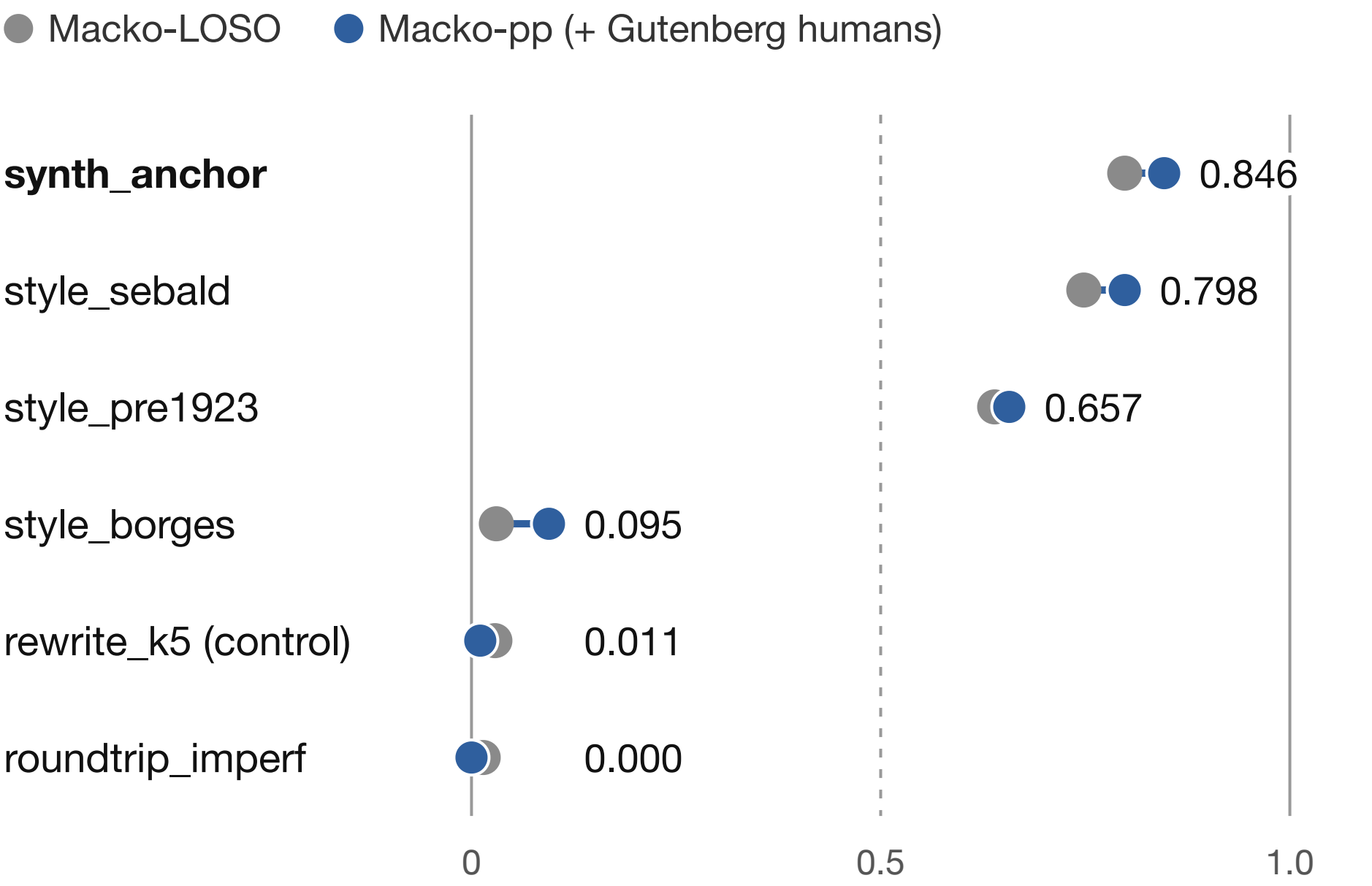


Figure 2. Fool rate on Macko-LOSO vs Macko-pp, Claude Opus 4.7, 20 held-out VK-2026 topics. Adding period prose to the human side does not close the cross-decade attack.

5. Competition results

Rank	Submission	Mean score ↓
1	uts-claude-v11	0.144
2	uts-claude-v4	0.164
3	roundtrip_imper	0.197
4	uts-claude-v7	0.211
5	uts-qwen-v7	0.308
6	Dargk	0.320

Table 3. Final ELOQUENT 2026 VK leaderboard (top 6). Our submissions in bold hold all five top positions. Rank 1 is an 8-detector minimax portfolio dominated by stream_narrative.

4. Results (continued)

4.4 Two detector postures, one Pareto frontier

Against a neutral PAN’25-trained committee ($n = 6,500$ Claude Opus candidates, 11 strategies), the ordering inverts: period-register attacks are caught (≤ 0.19) while register reduction (caveman) and literary mimicry win. No strategy in that pool beats both classes. Forcing a named **stream-of-consciousness witness** onto every topic (stream_narrative, $n = 300$) does: ≥ 0.97 on the neutral committee and 0.65 on Macko-pp.

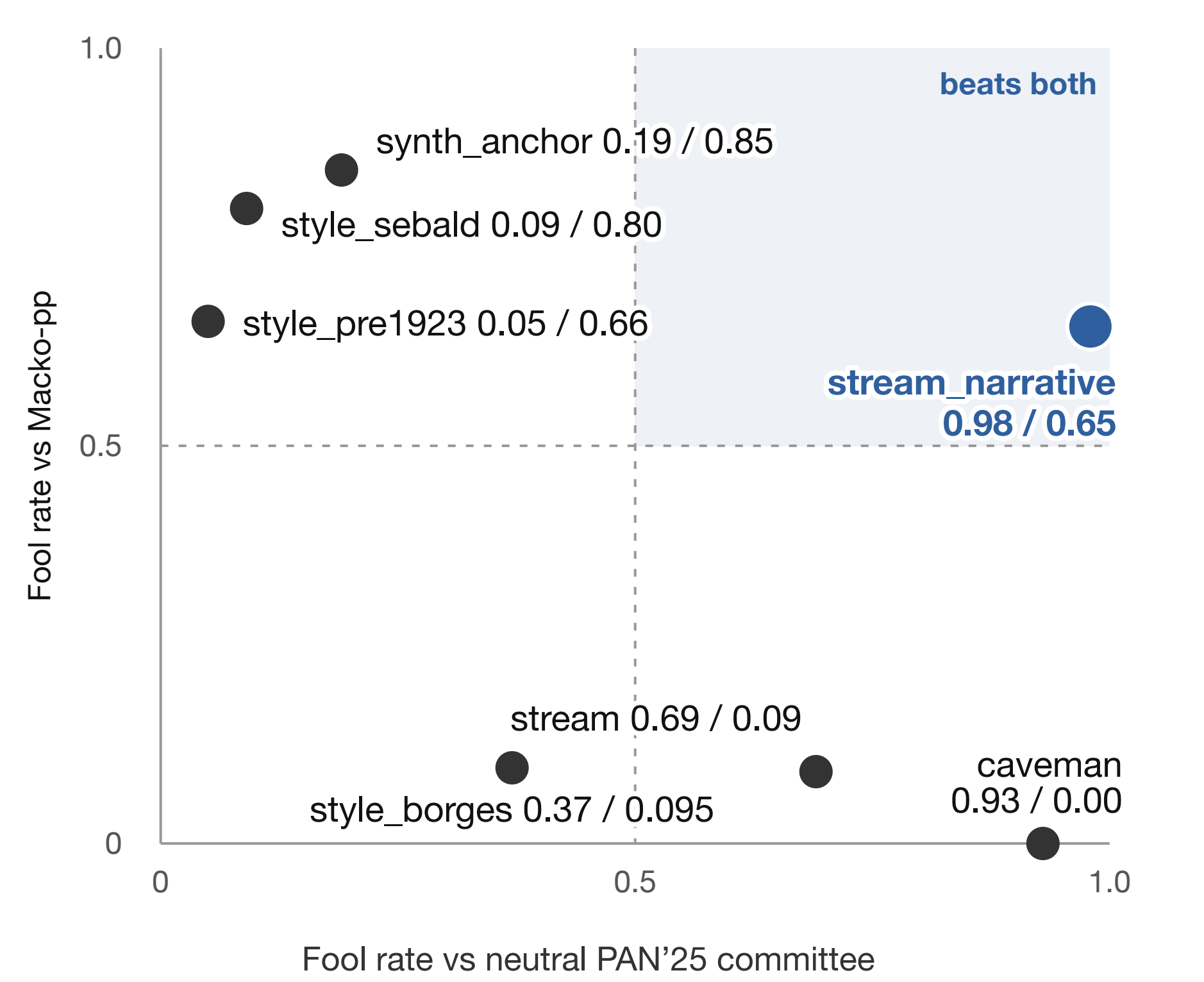


Figure 3. Each point is a strategy (neutral-committee fool rate / Macko-pp fool rate). Only stream_narrative lies in the upper-right region.

4.5 No naturalness penalty

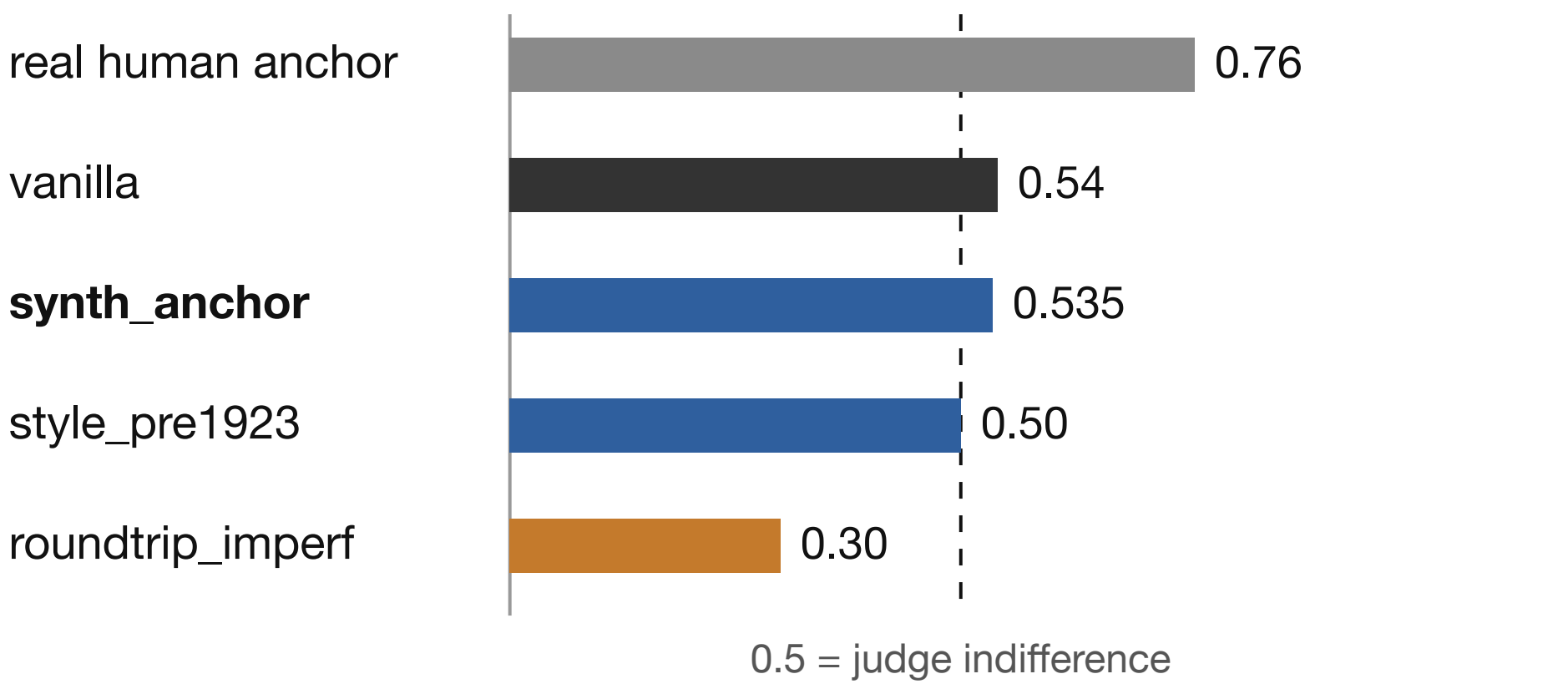


Figure 4. Cross-judge $P(\text{human})$, Claude Opus 4.7, $n = 40$ per cell. PJS v2 pays for evasion with naturalness ($0.54 \rightarrow 0.30$); the cross-decade attacks recover it while delivering the highest Macko-LOSO fool rates.

6. Conclusions

- Superficial evasion (grammar-level perturbation, the 2025 recipes) is closable inside one adversarial fine-tune: fool rate ≤ 0.025 .
- Structural shifts in register and form survive: 0.846 (period register) and 0.65 (stream narrative) on the hardened detector. The register attack carries no naturalness penalty.
- Augmenting training data with real period prose does not close the vulnerability; it widens the “human” region without separating human from emulated period prose.
- For the detector families tested, the operative variable is out-of-distribution direction, not surface mimicry: pulling text into the training distribution (PAN mimicry) fails at 0.000.
- Prompting alone suffices: no retraining, no model access, as few as two API calls per topic (synth_anchor).

Artifacts. Detector checkpoints and attack prompts released: huggingface.co/protagonist/eloquent26-macko-replica and the PAN’25 baseline [protagonist/roberta-eloquent](https://huggingface.co/protagonist/roberta-eloquent).

References

- [1] Bevendorff et al. Overview of the Voight-Kampff Generative AI Authorship Verification Task at PAN and ELOQUENT 2024. CLEF 2024.
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- [5] Hans et al. Spotting LLMs with Binoculars: Zero-Shot Detection of Machine-Generated Text. ICML 2024.
- [6] Macko, Moro, Srba. mdok of KInIt: Robustly Fine-tuned LLM for Binary and Multiclass AI-Generated Text Detection. CLEF 2025.
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